- In this architecture, each device has its own private key and associated public key.
- The public key is packaged into a certificate that is signed by a certificate authority (CA) and each device has a permanent copy of the public key of the CA.
- When a device connects to another device (or receives a connection), it presents its certificate to identify itself. The device authenticates with the associated private key to prove that it is the rightful holder of this certificate.
- The recipient can verify the identity of the other device by checking that its certificate is signed by the trusted CA and has not expired, been revoked, or in any other way become invalid.

This architecture can also be used in IoT environments and is often used for more capable devices. But constrained devices that lack the capacity for public key cryptography are unable to make use of a PKI and so other alternatives must be used, based on symmetric cryptography. Symmetric cryptography is efficient but requires the API client and server to have access to the same key, which can be a challenge if there are a large number of devices involved. The key distribution techniques described in the next few sections aim to solve this problem.

12.4.1 One-off key provisioning

The simplest approach is to provide each device with a key at the time of device manufacture or at a later stage when a batch of devices is initially acquired by an organization. One or more keys are generated securely and then permanently